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Perspective | [Open access](#) | Published: 13 March 2026

A transdiagnostic model for how general purpose AI chatbots can perpetuate OCD and anxiety disorders

[Ashleigh Golden](#) & [Elias Aboujaoude](#) 

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Data availability

No datasets were generated or analysed during the current study.

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4. Yankouskaya, A., Liebherr, M. & Ali, R. Can ChatGPT be addictive? A call to examine the shift from support to dependence in AI conversational large language models. *Hum.-Cent. Intell. Syst.* (2025). <https://doi.org/10.1007/s44230-025-00090-w>
5. Heinz, M. V. et al. Randomized trial of a generative AI chatbot for mental health treatment. *NEJM AI* 2, A10a2400802 (2025). <https://doi.org/10.1056/A10a2400802>
6. Hill, K. They asked A. I. chatbots questions. The answers sent them spiraling. *The New York Times* (13 June 2025). Available at: <https://www.nytimes.com/2025/06/13/technology/chatgpt-ai-chatbots-conspiracies.html>

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10. Klee, M. ChatGPT lured him down a philosophical rabbit hole. Then he had to find a way out. Rolling Stone (10 August 2025). Available at:
<https://www.rollingstone.com/culture/culture-features/chatgpt-philosophical-rabbit-hole-spiral-1234567890/>
11. Klee, M. Grieving parents tell Congress that AI chatbots encouraged their children to self-harm. Rolling Stone (17 September 2025). Available at:
<https://www.rollingstone.com/politics/politics-news/ai-chatbots-congress-hearing-self-harm-1234567891/>
12. Roose, K. Can A. I. be blamed for a teen's suicide? The New York Times (23 October 2024). Available at: <https://www.nytimes.com/2024/10/23/technology/characterai->

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17. AdBitter3655. Please stop using ChatGPT/AI for reassurance. Reddit (2025).

Available at:

https://www.reddit.com/r/OCD/comments/1hve8k3/please_stop_using_chatgptai_for_reassurance/

18. OneFish2Fish3. Does anyone else use ChatGPT for reassurance checking? *Reddit* (2024). Available at:

https://www.reddit.com/r/OCD/comments/1glihxp/does_anyone_else_use_chatgpt_for_reassurance/

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22. Pugachevsky, J. The rise of ChatGPT therapy and our constant need for feedback. Business Insider (17 March 2025). Available at:
<https://www.businessinsider.com/chatgpt-therapy-risks-benefits-boundaries-2025-3>
-
23. Duong, C. D., Dao, T. T., Vu, T. N., Ngo, T. V. N. & Tran, Q. Y. Compulsive ChatGPT usage, anxiety, burnout, and sleep disturbance: a serial mediation model based on stimulus–organism–response perspective. *Acta Psychol. (Amst.)* **251**, 104622.
<https://doi.org/10.1016/j.actpsy.2024.104622> (2024).
-
24. Zhong, W., Luo, J. & Lyu, Y. How do personal attributes shape AI dependency in Chinese higher education context? Insights from needs frustration perspective. *PLoS One* **19**, e0313314, <https://doi.org/10.1371/journal.pone.0313314> (2024).

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and treatment. *J. Consult. Clin. Psychol.* **64**, 1152–1168, <https://doi.org/10.1037/0022-006X.64.6.1152> (1996).

-
29. Starcevic, V. et al. Functions of compulsions in obsessive–compulsive disorder. *Aust. N. Z. J. Psychiatry* **45**, 449–457, <https://doi.org/10.3109/00048674.2011.567243> (2011).
-
30. Borkovec, T. D., Alcaine, O. M. & Behar, E. Avoidance theory of worry and generalized anxiety disorder. In *Generalized Anxiety Disorder: Advances in Research and Practice* (eds Heimberg, R. G., Turk, C. L. & Mennin, D. S.) 77–108 (The Guilford Press, 2004).

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34. Morrin, H. et al. Delusions by design? How everyday AIs might be fuelling psychosis (and what can be done about it). OSF Preprints (2025).
https://doi.org/10.31234/osf.io/cmy7n_v5
-
35. Sun, N. Y., Pittenger, C. & Ching, T. Analyzing the contents of a large, public online peer support forum for obsessive-compulsive disorder: thematic analysis. *JMIR Form. Res.* **9**, e60899, <https://doi.org/10.2196/60899> (2025).
-
36. Song, Y. et al. The effect of exposure and response prevention therapy on obsessive-compulsive disorder: a systematic review and meta-analysis. *Psychiatry Res.* **317**, 114861. <https://doi.org/10.1016/j.psychres.2022.114861> (2022).

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40. Grayson, J. *Freedom from Obsessive Compulsive Disorder: A Personalized Recovery Program for Living with Uncertainty* (Penguin, 2014).
-
41. International Medical Device Regulators Forum. Software as a medical device (SaMD): Key definitions. IMDRF SaMD Working Group (9 December 2013). Available at: <https://www.imdrf.org/docs/imdrf/final/technical/imdrf-tech-131209-samd-key-definitions-140901.pdf>
-
42. Illinois Department of Financial and Professional Regulation. Gov Pritzker signs legislation prohibiting AI therapy in Illinois. Press release (4 August 2025). Available at: <https://idfpr.illinois.gov/news/2025/gov-pritzker-signs-state-leg-prohibiting-ai-therapy-in-il.html>

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46. Golden, A. & Aboujaoude, E. Describing the framework for AI tool assessment in mental health and applying it to a generative AI obsessive-compulsive disorder platform: tutorial. *JMIR Form. Res.* **8**, e62963, <https://doi.org/10.2196/62963> (2024).

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Contributions

A.G. wrote the first draft of the manuscript. A.G. and E.A. both participated in background

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